

RailGuard

Autonomous Quadraped Robot and Handheld Security System for Smarter & Safer Railways

Technology Details for Narcotics andExplosives Detection in Indian Railways

Purpose. RailGuard is a technology blueprint for a rugged mobile quadraped robot and companion handheld device for RPF screening, surveillance, inspection and incident alerting. It is intended for concept design, prototype development and a Smart India Hackathon-style project proposal.

Safety position. RailGuard is a screening, inspection and decision-support platform. A detection alert must be verified by trained RPF, forensic or bomb-disposal personnel under approved SOPs. The robot must not autonomously touch, manipulate or neutralise a suspicious object.

1. Solution at a Glance

Subsystem	Recommended Implementation	Operational Role
Robot software	Ubuntu Linux + ROS 2 Humble/Jazzy	Sensor integration, robot control, navigation and mission management
Navigation	LiDAR SLAM, Nav2, IMU and camera/leg-odometry fusion	GPS-denied mapping, obstacle avoidance, patrol routes and safe return
Edge AI	NVIDIA Jetson Orin NX/AGX Orin or industrial equivalent	Real-time vision, thermal analysis, sensor fusion and local alerts
Chemical detection	IMS + Raman or NIR; e-nose as a screening layer	Trace alerts and material-screening workflow
Vision	RGB, low-light and radiometric thermal cameras	Surveillance, unattended-baggage alerts, inspection and night patrol
Connectivity	4G/5G, Wi-Fi 6 and secured mesh-radio backup	Live video, telemetry, remote control and control-centre alerts
Backend	MQTT/DDS, FastAPI, PostgreSQL/PostGIS, React dashboard	Fleet monitoring, incident records, map view, audit trail and reporting
Handheld device	Rugged ARM system with IMS and Raman/NIR	Rapid personnel-operated screening, offline logging and synchronisation

2. Programming Languages and Frameworks

Area	Technology	use
Real-time robotics	C++17/20, ROS 2, rclcpp	Low-latency sensor drivers, control nodes, navigation and safety interfaces
Application logic	Python 3, rclpy	Rapid prototyping, data processing, AI orchestration and test scripts
AI and vision	PyTorch, OpenCV, ONNX Runtime, TensorRT, CUDA	Training, detection, tracking, segmentation and accelerated edge inference
Navigation	Nav2, SLAM Toolbox, robot_localization	Path planning, localisation, mapping and sensor fusion
Simulation	Gazebo/Ignition or NVIDIA Isaac Sim	Virtual railway-yard/coach scenarios and regression testing

Area	Technology	Use
Web dashboard	React, TypeScript, MapLibre/Leaflet	Live control-room interface, map, incidents and fleet status
Backend/API	FastAPI or Node.js, REST, WebSocket	Device API, alert workflows, user management and dashboard services
Messaging	MQTT and/or ROS 2 DDS	Efficient telemetry, alerts, health data and command messaging
Data	SQLite on device; PostgreSQL/PostGIS centrally	Offline event storage, geospatial queries, reporting and retention
DevOps	Git, Docker, CI/CD, unit/integration tests	Repeatable builds, version control and secure deployment

3. Quadruped Hardware Architecture

3.1 Rugged mobility platform

- Fourindependentlyactuated legsusing brushless motors, joint encoders and torque/current feedback.
- High-clearance chassis and protected lower body for uneven ballast, yard terrain, cables and coach underframe access.
- Operating modes: stand, walk, crawl, crouch, precision inspection, teleoperation, return-to-base and emergency stop.
- Ingress-protected enclosure, vibration isolation, protected sensor windows and corrosion-resistant fasteners.
- Hot-swappable battery packs with battery-management system, thermal monitoring and low-battery safe-return logic.

Independent hardwired emergency stop, remote stop command and watchdog-controlled safe shutdown.

3.2 Compute and control

Component	Recommended Specification	Purpose
Primary edge computer	Jetson Orin NX for prototype; AGX Orin/industrial GPU computer for high payload	Runs AI, video processing, sensor fusion, SLAM and mission control
Safety MCU	STM32 or safety-oriented TI/automotive-class controller	Motor interlocks, e-stop, battery monitoring and independent watchdog
Memory/storage	16–32 GB RAM; 512 GB–2 TB industrial NVMe SSD	Models, maps, encrypted logs and locally buffered video
Power	Isolated DC-DC rails and fused power distribution	Separates sensitive compute/sensors from actuator noise
Time source	RTC plus NTP/PTP; GNSS time outdoors	Accurate incident timestamps and sensor-data alignment

4. Sensor Payload

Sensor	Technology / Interface	Primary Use	Important Limitation
3D LiDAR	360-degree 3D LiDAR; Ethernet/UDP	SLAM, obstacle detection, mapping and route planning	Performance can degrade in heavy rain, dense dust or reflective surfaces
IMU and encoders	9-axis IMU, joint encoders, CAN/Ethernet	Attitude estimation, odometry and stable walking	Requires calibration and fusion with other sensors

Sensor	Technology / Interface	Primary Use	Important Limitation
RGB cameras	Wide-angle/4K, global shutter preferred	Visual patrol, package/person detection, teleoperation	Low-light performance varies
Thermal camera	Radiometric thermal imager	Night visibility, heat anomalies, fire/smoke support	Does not identify narcotics or explosives chemically
IMS	Swab and/or vapour trace-detection module	Trace screening for explosives and narcotics	Requires approved sampling, calibration and consumables
Raman	Portable Raman spectrometer on pan-tilt mount	Material screening/identification where optical access exists	Opaque packaging, fluorescence and contamination may limit results
NIR (optional)	Near-infrared spectrometer	Bulk material screening through some packaging	Not a substitute for trace detection or forensic confirmation
E-nose	Gas-sensor array, pump, humidity/temperature compensation	Broad anomaly indication and prioritised inspection	Not definitive substance identification
Metal detector	EMI/magnetic anomaly sensor	Hidden metal, wiring and metallic anomalies	Cannot detect non-metallic threats alone
GNSS/UWB	GNSS outdoors; UWB/beacons indoors	Event tagging and improved indoor positioning	GNSS unavailable in tunnels and many covered areas

5. AI, Analytics and Decision Support

5.1 Vision and thermal analytics

- Object detection for people, luggage, parcels, vehicles, track obstacles and restricted-zone intrusions.
- Multi-object tracking to identify unattended baggage using location, owner association and dwell-time logic.
- Underframe/coach inspection using image comparison and anomaly-detection models trained on approved inspection imagery.
- Thermal anomaly detection for unusual heating, fire/smoke indicators and night-time situational awareness.
- Optical character recognition for coach identifiers, parcel labels and station-area markers where permitted.
- Facial matching should be optional, access-controlled, legally authorised and operated only against approved watchlists.

5.2 Chemical-sensor fusion

Chemical output should be treated as confidence-scored alerts. The fusion engine combines IMS peak information, Raman/NIR spectra, e-nose anomaly scores, location, sampling method and sensor-health status. It should explicitly support inconclusive and repeat sample outcomes rather than forcing a positive/negative declaration.

Output Field	Example
Result category	Suspected explosive; suspected narcotic; no alert; inconclusive
Confidence	Low / medium / high with a calibrated threshold
Evidence source	IMS swab, Raman scan, NIR scan, e-nose anomaly or combined result
Context	Robot/device ID, operator ID, location, timestamp, camera clip reference
Required action	Rescan, isolate area, supervisor review, EOD/RPF escalation

6. Navigation and Mission Software

RailGuard should run a ROS 2 graph in which each sensor publishes time-stamped data, a localisation layer produces the robot pose, and Nav2 plans and controls a safe path. LiDAR, IMU, visual odometry and leg/joint odometry should be fused using an Extended Kalman Filter or factor-graph approach.

ROS 2 Module	Function
Sensor driver nodes	Acquire LiDAR, cameras, thermal imagery, IMU, GNSS, IMS/Raman status and motor telemetry
robot_state_publisher	Publishes robot body and leg transforms
robot_localization	Fuses IMU, visual, LiDAR and leg-odometry data
slam_toolbox	Builds and updates a map in GPS-denied station, tunnel and yard environments
Nav2 planner/controller/costmaps	Plans routes, avoids obstacles and manages motion constraints
Quadruped gait controller	Converts body-motion commands into leg motions while enforcing stability limits
Mission manager	Runs patrol, inspection, return-to-base and alert-response workflows
Safety supervisor	Monitors e-stop, tilt, collision, lost-communications, low-battery and fault conditions

7. Handheld Detection Device

Area	Recommended Details
Form factor	Target below 1.5 kg; drop-resistant, weather-resistant, sunlight-readable 5–7 inch touch display
Core sensing	IMS for trace swab/vapour screening; Raman or NIR for material screening; temperature/humidity monitoring
Processing	Industrial ARM processor with secure element; optional small AI accelerator
User alerts	Buzzer, vibration, high-visibility LED, text result and confidence indicator
Connectivity	4G/5G, Wi-Fi, Bluetooth Low Energy; secure store-and-forward when disconnected
Offline operation	Encrypted local database, timestamping, device/operator ID and automatic sync when a network returns
Languages	Hindi and English first; modular localisation for regional languages
Audit trail	Scan ID, sample type, location, operator acknowledgement, calibration state and chain-of-custody fields

8. Communications, Dashboard and Data

8.1 Communications

- Primary communications: encrypted 4G/5G or private LTE; local station Wi-Fi 6 where authorised.
- Fallback communications: secured mesh radio for commands, health telemetry and emergency alerts.
- Use adaptive video streaming: preserve control and alerts first, then reduce video bitrate during poor connectivity.
- Use store-and-forward queues so incidents are not lost during tunnel, yard or network outages.

8.2 Control-centre dashboard

- Live position of robot and handheld assets on station, yard or geographic maps.
- Live RGB and thermal streams with device health, battery, link quality and sensor-calibration state.

- Alert queue showing incident level, confidence, sensor evidence, time, location, operator and recommended escalation.
- Searchable encrypted incident history, patrol replay, maintenance schedule and exportable incident reports.
- Role-based views for field officer, station supervisor, divisional officer and system administrator.

9. Cybersecurity, Privacy and Safety

Control	Implementation
Secure device identity	Hardware-rooted identity, secure boot and signed firmware/model packages
Data protection	AES-256 encryption at rest; TLS 1.3/mutual authentication in transit
Access control	Role-based access, strong authentication, audit logs and no default credentials
Network separation	Separate control, video, chemical-alert and administrative networks
Privacy controls	Purpose limitation, retention policy, access audit and legally approved workflow for biometric data
Functional safety	Physical e-stop, remote stop, collision avoidance, speed limits near public areas and fall detection
Fail-safe behaviour	Stop/hold, return to safe point or controlled shutdown after critical fault, low battery or loss of control link
Incident handling	No autonomous evidence handling; trained RPF/EOD staff verify and manage suspicious items

10. Prototype Build Plan

Phase	Work Package	Main Deliverable
1. Core platform	Mobility integration, e-stop, power management and teleoperation	Safe controllable quadruped base
2. Navigation	LiDAR, IMU, cameras, mapping and obstacle avoidance	Indoor/outdoor GPS-denied patrol demonstration
3. Perception	RGB/thermal analytics and event logging	Unattended-object, intrusion and anomaly demonstration
4. Chemical interface	Integrate approved commercial/lab IMS and Raman/NIR interface	Sensor health, scan workflow and confidence alerting
5. Handheld	Rugged handheld UI, offline store and synchronisation	Field-operable screening workflow
6. Backend	Encrypted messaging, dashboard and role-based access	Central alert and fleet-management platform
7. Validation	Dust, heat, vibration, low-light, outage and battery tests	Test evidence and deployment readiness report

11. Recommended Prototype Bill of Materials

Item	Prototype Recommendation	Notes
Quadruped base	Commercial research/developer platform or indigenous actuator prototype	Prefer a platform with documented SDK, payload capacity and e-stop
Edge computer	Jetson Orin NX development carrier	Scale to AGX Orin/industrial system for multi-camera production configuration
LiDAR	One 3D top-mounted LiDAR plus short-range coverage	Choose industrial-grade interface and environmental sealing

Item	Prototype Recommendation	Notes
Vision	Front RGB, side/rear cameras, low-light camera and thermal unit	Synchronise timestamps where possible
Chemical module	Commercial IMS and Raman/NIR interface	Use certified reference/testing arrangements; do not manufacture or handle prohibited materials
Network	4G/5G router, Wi-Fi 6, radio fallback	Use external, protected antennas
Power	Hot-swap battery, BMS, isolated DC-DC rails	Size power budget for peak actuator load and sensor payload
Mechanical	Aluminium/composite enclosure, vibration mounts, mast and pan-tilt	Maintain serviceability and safe centre of gravity

12. Testing and Acceptance Criteria

- **Navigation:** autonomous mapping and supervised patrol in a GPS-denied indoor test route; safe stop for people and obstacles.
- **Mobility:** stable movement on representative railway-like uneven terrain without sensor-mast instability.
- **Perception:** measurable detection/false-alarm metrics for packages, intrusion and low-light conditions.
- **Chemical workflow:** calibration, blank checks, known safe reference materials, repeatability and documented false-alarm testing by authorised experts.
- **Communications:** alert delivery with delayed/offline queue recovery; no loss of audit records after a network interruption.
- **Cybersecurity:** signed updates, credential management, encrypted storage and access-log verification.
- **Operational usability:** RPF-oriented UI trial with short training, clear alarms and documented escalation SOP.

13. Final Recommended Architecture

Robot layer: Rugged quadruped + Jetson edge computer + ROS2 + LiDAR/IMU/camera fusion + Nav2/SLAM + RGB/thermal surveillance + IMS/Raman/NIR sensor interfaces.

Handheld layer: Rugged encrypted field device + IMS and Raman/NIR workflow + offline-first records + multilingual alert interface.

Command layer: MQTT/DDS telemetry + FastAPI/Node services + PostgreSQL/PostGIS + React dashboard + encrypted video/event storage.

Safety layer: Human-supervised operations, independent emergency stop, confidence-scored chemical alerts, cybersecurity controls and RPF/EOD verification before response actions.

References and Technical Basis

The design uses ROS2 Nav2 and SLAM Toolbox for mapping/navigation; NVIDIA Jetson Orin for edge robotics AI; and IMS with Raman/NIR as complementary field-screening approaches for explosives and narcotics. Final production selection must be validated through accredited laboratory testing, environmental trials and Railway/RPF procurement and safety approvals.